

A fixed representative family of CAR shell data

MathlibAnnex.CStarAlgebra.CAR.RepresentativeShellFamily

structure

One family records a transporting automorphism and exact shell links for each selected pure-state class, with an explicitly fixed root component.

Statement

Let A be the completed CAR algebra, the norm completion of the matrix stages $A_n = M_{2^n}(\mathbb{C})$ under $a \mapsto a \otimes I_2$, with the fixed coordinate reindexing. Write $\iota_n : A_n \rightarrow A$ for the canonical inclusion, $\omega(\iota_n(a)) = a_{00}$ for the root state, $f_n = \iota_n(E_{00})$ for its decreasing root projections, and $p_n = f_n - f_{n+1}$ for the difference shells. Each p_n is a self-adjoint projection. Let I be the set of unitary-equivalence classes of pure-state Gelfand–Naimark–Segal (GNS) representations of A . Choose a representative pure state φ_j for each $j \in I$, with the distinguished class $o = [\omega]$ represented by $\varphi_o = \omega$. A representative shell family assigns to each $j \in I$ an automorphism α_j and links $w_{j,n} \in A$ with $\varphi_j(\alpha_j(a)) = \omega(a)$ for every a , $w_{j,n}^* w_{j,n} = \alpha_j(p_n)$, and $w_{j,n} w_{j,n}^* = p_n$ for every n . At the distinguished class it further requires $\alpha_o = \text{id}_A$ and $w_{o,n} = p_n$.

Definition

The component structure stores α_j , the equation $\varphi_j \circ \alpha_j = \omega$, the sequence $(w_{j,n})$, and its initial and final supports. The family structure chooses a component for every class and adds the root equations. They normalize a distinguished component of this chosen family; they do not impose equality between arbitrary choices made in unrelated existence statements.

Assumptions

The CAR system, root state, and selected representatives are fixed. A family is supplied as data of this structure. There is no restriction that I be countable, and this structure declaration does not itself prove the existence of such a family.

Conclusion

The fields are the component data for every class and the two root equations. Each component contains one automorphism, its state-transport equation, its link sequence, and both support equations. The root shell $p_n = f_n - f_{n+1}$ is distinguished from the nested flag projection f_n .

No rank-one, capture, simplicity, compactness, or approximate-innerness conclusion is stored in this structure. A later constructor can use approximate innerness to build the links, but that construction property is not an extra field required of every supplied family.

Main citations

- The stated existence or structural result —
MathlibAnnex.CStarAlgebra.CAR.RepresentativeShellFamily
- Exact component fields — MathlibAnnex.CStarAlgebra.CAR.RepresentativeShellData
- Difference shell definition — MathlibAnnex.CStarAlgebra.CAR.rootShell
- Projection property of each difference shell —
MathlibAnnex.CStarAlgebra.CAR.isStarProjection_rootShell
- The identity component satisfies the required equations —
MathlibAnnex.CStarAlgebra.CAR.rootShell_identity_family

Supporting route explanation

Lean source signature (exact)

```
structure RepresentativeShellFamily where
  data : ∀ j : MathlibAnnex.CStarAlgebra.PureState.GNSClass Limit, RepresentativeShellData j
  root_alpha : (data completedRootPureState.classOf).alpha =
    StarAlgEquiv.refl Ⓢ Limit
  root_link : ∀ n, (data completedRootPureState.classOf).link n = rootShell n
```

In the source	Mathematical meaning
GNSClass Limit	The set I of unitary-equivalence classes of pure-state GNS representations of the CAR algebra A .
completedRootPureState; completedRootPureState.classOf	The root state ω , bundled with purity, and its class $o = [\omega] \in I$.
data : ∀ j : ... GNSClass Limit, RepresentativeShellData j	For each $j \in I$, one component containing α_j , the state equation, the links $w_{j,n}$ and both support identities. These five component fields are displayed separately below.
root_alpha : (data completedRootPureState.classOf).alpha = StarAlgEquiv.refl Ⓢ Limit	The distinguished component must have $\alpha_o = \text{id}_A$.
root_link : ∀ n, (data completedRootPureState.classOf).link n = rootShell n	For every n , the same distinguished component must have $w_{o,n} = p_n = f_n - f_{n+1}$, where $f_n = \iota_n(E_{00})$. The shell is not the flag projection f_n .

Separate related structure, RepresentativeShellData j (the same fixed source, lines 28–37):

```
structure RepresentativeShellData (j : MathlibAnnex.CStarAlgebra.PureState.GNSClass Limit)
where
  alpha : Limit  $\simeq_{\star_a} [\mathbb{C}]$  Limit
  state_eq : ∀ a : Limit,
    (MathlibAnnex.CStarAlgebra.PureState.representative completedRootPureState j).1 (alpha a) =
```

```

    completedRootPureState.1 a
  link : N → Limit
  initial_support : ∀ n,
    star (link n) * link n = alpha (rootShell n)
  final_support : ∀ n,
    link n * star (link n) = rootShell n

```

In the source	Mathematical meaning
<code>alpha : Limit $\simeq_{\mathbf{a}}$ [C] Limit</code>	The automorphism $\alpha_j : A \rightarrow A$ for the specified class j .
<code>state_eq : ∀ a : Limit, (...) .1 (alpha a) = completedRootPureState.1 a</code>	For every $a \in A$, $\varphi_j(\alpha_j(a)) = \omega(a)$. The <code>.1</code> values here are the underlying state functionals of the selected pure-state representatives.
<code>link : N → Limit</code>	The sequence $n \mapsto w_{j,n} \in A$ for this same α_j .
<code>initial_support : ∀ n, star (link n) * link n = alpha (rootShell n)</code>	For every n , $w_{j,n}^* w_{j,n} = \alpha_j(p_n)$.
<code>final_support : ∀ n, link n * star (link n) = rootShell n</code>	For every n , $w_{j,n} w_{j,n}^* = p_n$, where $p_n = f_n - f_{n+1}$. This is the final support, distinct from the transported initial support.

Further source notes: Here `completedRootPureState` is the root state bundled with its purity proof. The separate exact excerpt above shows the component's full fields.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.CStarAlgebra.CAR.RepresentativeShellFamily`

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